SUSTAINABLE TEXTILE PROCUREMENT AT

TU BERLIN

From reuse concept to evidence-based circular textile management

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Closing the Loop for Used Textiles on Campus is a circular textile management project developed within the Sustain.ALL Transformation HUB at TU Berlin. It explores how used institutional textiles such as workwear, laboratory coats and safety shoes can be collected, sorted and directed toward suitable end-of-life pathways. The project also aims to connect procurement decisions with more effective textile recovery and recycling.

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Aishwarya Vithal Kharat
Catherine Cynthia Chrisanta
Rachel / Xiaorui Dong

1 Project overview

Closing the Loop for Used Textiles on Campus is a circular textile management project developed within the Sustain.ALL Transformation HUB at TU Berlin. It explores how institutional textiles such as workwear, laboratory coats and safety shoes can be collected, assessed and directed toward suitable end-of-life pathways.

The project initially focused on a central reuse-oriented textile hub. Stakeholder research later showed that direct reuse is limited for many textile streams because of wear, contamination and material complexity. The concept therefore evolved into a broader system for textile collection, sorting, and routing.

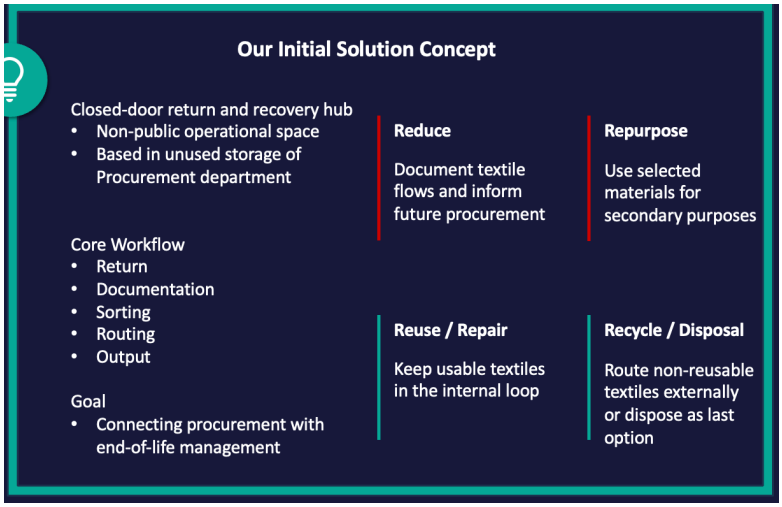


Figure 1: Initial concept for a central textile return and routing hub at TU Berlin.

2 Project partners & Team

The project was developed within the Sustain.ALL Transformation HUB at TU Berlin. The student team worked with relevant TU Berlin departments to understand current textile procurement, return processes, and end-of-life practices.

Name	Affiliatioin/Programme	Role
Aishwarya Vithal Kharat	B.Sc. Nachhaltiges Management	Project team
Catherine Cynthia	M.Sc. Environmental	Project team

Chrisanta	Science & Technology	
Rachel / Xiaorui Dong	M.Sc. Industrial Engineering	Project team
Central Procurement Department	TU Berlin	Stakeholder: procurement data, textile return process and purchasing practices

3 Project Objectives

TU Berlin uses different types of institutional textiles, including workwear, laboratory coats, safety shoes and apprentice clothing. However, procurement, return and end-of-life management are not fully centralized, and information about textile quantities, material composition and existing disposal pathways is limited.

The project initially focused on the reuse or resale of returned textiles. Stakeholder feedback showed that this is not realistic for many textile streams because of wear, contamination, hygiene requirements and mixed materials.

The project therefore reframed its central question from:

> “How can we reuse or resell used textiles at TU Berlin?”

to:

> “How can TU Berlin safely and efficiently collect, sort and route used textiles to the most appropriate circular pathway?”

The main goals of the project were to:

- understand existing textile flows and return processes
- identify realistic end-of-life pathways
- investigate recycling and supplier take-back options
- identify potential external partners
- connect procurement decisions with end-of-life management

In the longer term, the project aims to support a more transparent system linking procurement, use, return, sorting and end-of-life treatment.

4 Research Process

The project combined system analysis, stakeholder interviews and external partner research. The findings were used to test the initial reuse-oriented concept and develop a more realistic approach for circular textile management.

4.1 Literature Research

The team examined available information on textile types, procurement, return processes and current end-of-life handling.

This analysis identified several gaps, particularly regarding textile quantities, material composition, responsibilities and existing recycling pathways.

4.2 Stakeholder Interview

Interviews with the Central Procurement Department and Facility Management helped clarify current procurement practices, textile return processes, storage possibilities and operational responsibilities.

The interviews showed that direct reuse is mainly realistic for unused or nearly new textiles. Workwear and laboratory clothing may be heavily worn or contaminated, while safety shoes are difficult to recycle because of their mixed-material construction.

4.3 External Partner Research

Several recycling and circular-economy organisations were contacted to investigate possible recycling, recovery and take-back solutions.

The research focused on accepted textile fractions, contamination requirements, sorting, minimum quantities, transport arrangements and costs.

Most organisations contacted did not respond. Re-Fresh Global was the only external partner that responded and provided a proposal for further testing.

The partner research showed that TU Berlin would require cooperation with specialised external organisations to establish suitable recycling and recovery pathways.

5 Stakeholder Insights and Feasibility Barriers

Stakeholder interviews were used to understand how used institutional textiles are currently managed at TU Berlin and which circular options are realistic.

The interviews showed that the reuse potential of most textiles is very limited. Workwear is often heavily worn or dirty after use. Laboratory coats may contain chemical residues, which creates hygiene and safety risks. Safety shoes are also difficult to recycle because they contain several different materials.

Key findings from the stakeholder interviews:

- Most returned workwear and laboratory clothing is not suitable for direct reuse.
- Reuse is mainly possible for textiles that are unused or almost new.
- Repair, recycling and disposal are considered more realistic pathways for most used textiles.
- TU Berlin introduced a return process in 2025, but only small quantities have been returned so far.
- Central Procurement currently manages textiles from the central university administration.
- Faculties often purchase and manage laboratory clothing independently.
- There is currently no university wide return and recycling system.
- No established recycling or take back partner is available.
- Storage capacity for a possible collection system is available at the Central Campus in Charlottenburg.

The stakeholder feedback showed that procurement data alone is not enough to evaluate circular potential. Textile condition, contamination, material composition and safety requirements also need to be considered.

6 Challenge Reframing

The initial project concept focused on the reuse and resale of used institutional textiles. However, stakeholder feedback showed that this approach did not reflect the actual condition of most textile streams at TU Berlin. Workwear and laboratory clothing are often heavily worn, dirty or potentially contaminated, while direct reuse is mainly realistic for unused or nearly new items.

The project focus therefore shifted from reuse towards safe and feasible end of life routing. Instead of asking how used textiles could be reused, the project considered how different textile streams could be collected, assessed and

directed to the most appropriate circular pathway.

The reframed challenge was defined as:

How can TU Berlin safely and efficiently collect, sort and route used textiles to the most appropriate circular pathway?

This reframing created the basis for further partner research and for the development of a textile routing framework that includes repair, recycling, downcycling and controlled disposal.

7 External Partner Validation

After the challenge was reframed, external partner outreach became the main project activity.

The purpose of the outreach was to identify companies that could process TU Berlin's used textile streams and to test whether recycling or downcycling could be implemented in practice.

Potential external partners included:

- Re Fresh Global
- Remondis
- HPC AG
- Bardusch
- SOEX I
- BTV Lohsa

The companies were contacted with information about approximately 300 kg of used institutional textiles.

The main questions included:

- Which textile fractions can be accepted?
- Can laboratory coats be recycled?
- Can safety shoes and other mixed material products be processed?
- Which sorting requirements apply?
- Are minimum quantities required?
- How could collection or transport be organized?

Only Re Fresh Global provided a direct response during the project period.

Re Fresh Global proposed a pilot using approximately 10 kg of representative

textile samples. The pilot would test recycling suitability and evaluate the quality of the recycled material.

- Pilot quantity: approximately 10 kg
- Estimated cost: €5,000 to €20,000
- Main open questions: project objective, budget, ownership, responsibilities and the future use of the recycled material.

The partner feedback showed that recycling is technically possible in principle. However, a pilot cannot be implemented without a clear objective, available budget and defined responsibilities.

The outreach therefore served as a practical feasibility check for the reframed project concept.

8 Circular Textile Routing Framework

The final project concept is a routing framework for used institutional textiles.

The framework aims to identify the safest and most suitable end of life pathway for each textile stream.

Basic routing logic:
Used textiles → Categorize material → Assess condition → Select end of life pathway

Possible pathways include repair, recycling, downcycling and controlled disposal.

The framework distinguishes three textile categories.

1. Non chemically contaminated textiles: Examples include T shirts, uniforms, trousers, pullovers and rain jackets. Possible pathways include repair, recycling and downcycling, depending on material composition and physical condition.
2. Chemically contaminated textiles: The main example is laboratory coats. Possible chemical residues can limit reuse and conventional recycling. These textiles may require specialized recycling or disposal partners.
3. Accessories and protective equipment: Examples include safety shoes, goggles and other protective equipment. These products often combine several materials and are therefore more difficult to recycle.

For most institutional textiles, recycling and downcycling were identified as the most realistic pathways.

Direct reuse has limited potential because many textiles are heavily worn, dirty or potentially contaminated. Used uniforms and laboratory clothing also have limited secondhand demand.

The long-term goal is to establish clear routing options for each textile category and identify suitable specialized partners.

A regular collection and processing process could then replace the current case by case handling of used textiles.

9 Sustainable Procurement Implications

The project showed that textile end of life management is strongly connected to procurement.

The circular potential of a textile is partly determined before the product is purchased. Material composition, product design, supplier information and available take back options can influence whether a textile can later be repaired, recycled or recovered.

Relevant implications for future procurement include:

- More detailed information about material composition
- Clearer information about recyclability
- Supplier information about available recycling options
- Supplier take back systems
- Consideration of end-of-life pathways during product selection
- Use of return and recycling data for future procurement decisions

The stakeholder interview showed that TU Berlin has not yet systematically asked suppliers about recycling or take back options. The interview also indicated that recycled content could potentially become a relevant procurement criterion if manufacturers provide reliable information in their product data.

The routing process can also provide feedback for procurement.

If a product is repeatedly difficult or expensive to recycle, this information can be considered during future purchasing decisions. Products with simpler material compositions or clearer recycling options may be more suitable from a circular economy perspective.

The project therefore connects two parts of the textile lifecycle:

Procurement → Use → Return → Routing → Procurement feedback

This feedback loop can help TU Berlin gradually improve material transparency, supplier cooperation and the circularity of future textile procurement.

10 Impact & Expected Outcomes

The project aims to create environmental, circular economy, and organizational impact by connecting TU Berlin's procurement decisions with the end-of-life management of institutional textiles. This approach reflects the EU's shift towards considering the entire lifecycle of textile products, including durability, repairability, recyclability and material recovery (European Commission, 2022).

The expected environmental impact is a reduction in textile waste and more efficient use of resources through reuse where feasible and recycling or downcycling where reuse is not possible. This is particularly relevant because textile consumption in the EU creates substantial pressures on water and land use and is among the highest consumption categories for raw-material use and greenhouse-gas emissions (European Environment Agency, 2026).

From a circular economy perspective, the project seeks to keep textile materials in productive use for as long as possible and recover their material value when they can no longer be used. However, stakeholder research showed that contamination, hygiene requirements and the condition of institutional textiles significantly limit direct reuse. Therefore, recycling and downcycling are considered realistic pathways for many of TU Berlin's used textiles.

The project can also create organizational impact by improving information about textile quantities, material composition and end-of-life pathways. It could support the Procurement Department in incorporating circularity considerations into future purchasing decisions and in establishing cooperation with specialized recycling partners. This corresponds with the EU's objective of making textiles more durable, repairable and recyclable and increasing producer responsibility along the value chain (European Commission, 2022).

11 SDG Alignment

The project contributes most directly to SDG 12 – Responsible Consumption and Production, which aims to ensure sustainable consumption and production patterns and includes the sustainable management and efficient use of natural resources (United Nations, n.d.).

The project supports SDG 12 by promoting sustainable textile procurement, reducing avoidable textile waste, improving material recovery and considering the entire lifecycle of products rather than focusing only on the purchasing stage.

It also contributes to SDG 13 – Climate Action. Textile production and consumption generate significant greenhouse-gas emissions, while reducing resource consumption, extending product lifetimes and increasing recycling can contribute to reducing environmental pressures associated with textiles (European Environment Agency, 2026).

SDG 9 –Industry, Innovation and Infrastructure is relevant because the project develops an innovative institutional system for textile collection, sorting and material recovery. SDG 9 explicitly includes promoting sustainable industrialization and fostering innovation (United Nations, n.d.).

Finally, SDG 17 – Partnerships for the Goals is particularly relevant because implementation requires cooperation between TU Berlin's Procurement Department, staff, suppliers and external recycling companies. The UN defines SDG 17 around strengthening partnerships for sustainable development (United Nations, n.d.).

Overall, SDG 12 is the project's primary SDG, while SDGs 9, 13 and 17 support the innovation, climate and collaboration dimensions of the project.

12 Project Milestones

The project developed through a series of research, validation and reframing stages.

1. Stakeholder Research & Needs Assessment

We conducted stakeholder interviews with the Procurement Department and Facility Management to understand existing textile streams, procurement processes, collection practices and end-of-life challenges.

2. Current System Analysis

We mapped the existing textile flows and identified fragmented procurement, use and return processes, as well as limited information about textile volumes, composition and current routing pathways.

3. Development of Initial Concept

Our initial concept focused on creating a circular return and reuse-oriented textile hub.

4. Reframing the Challenge

Stakeholder feedback showed that direct reuse is not feasible for many institutional textiles because of contamination, hygiene concerns and limited second-hand demand. We therefore shifted towards safe and feasible end-of-life routing, with recycling and downcycling becoming important pathways.

5. Development of Routing Framework

We categorized textiles according to their characteristics, including non-chemically contaminated textiles, chemically contaminated textiles and accessories/protective equipment, and considered different end-of-life pathways for each category.

6. External Partner Outreach

We identified multiple potential recycling and circular-economy partners and contacted several organizations with information about approximately 300 kg of used textiles.

7. Industry Validation

Re-Fresh Global responded with a proposal for a 10 kg representative-material pilot to test recycling suitability and the quality of recovered materials. The indicative cost was €5,000–€20,000.

8. Recommendations & Next Steps

The next stage would be to clarify the objective, budget and ownership of a pilot, validate the technical feasibility of the routing system and establish suitable long-term recycling partnerships.

13 Conclusion

The project began with the assumption that the main opportunity for TU Berlin was to reuse or resell used textiles. However, stakeholder interviews and external market research showed that this approach is not suitable for many institutional textiles because of contamination, hygiene requirements, material composition and limited demand for used workwear and laboratory clothing.

As a result, we reframed the challenge from “How can we reuse textiles?” to “How can TU Berlin safely and feasibly route used textiles to their most appropriate next pathway?” This reframing was one of the most important outcomes of the project.

Our findings suggest that closing the textile loop at TU Berlin requires better data, clearer responsibilities, appropriate collection and sorting processes, and cooperation with specialized external partners. The European Commission's Sustainable and Circular Textiles Strategy similarly emphasizes lifecycle thinking, recyclability, repairability, recycled materials and responsibility across the textile value chain (European Commission, 2022).

The response from Re-Fresh Global demonstrated that external recycling solutions are technically worth investigating, while also highlighting that a pilot requires a clear objective, budget and ownership.

Ultimately, our project did not produce a single “perfect” recycling solution.

Instead, it developed a stakeholder informed framework for making better textile decisions from procurement through end-of-life. The next logical step would be to validate this framework through a pilot and use the results to develop a scalable circular textile management system for TU Berlin.

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